

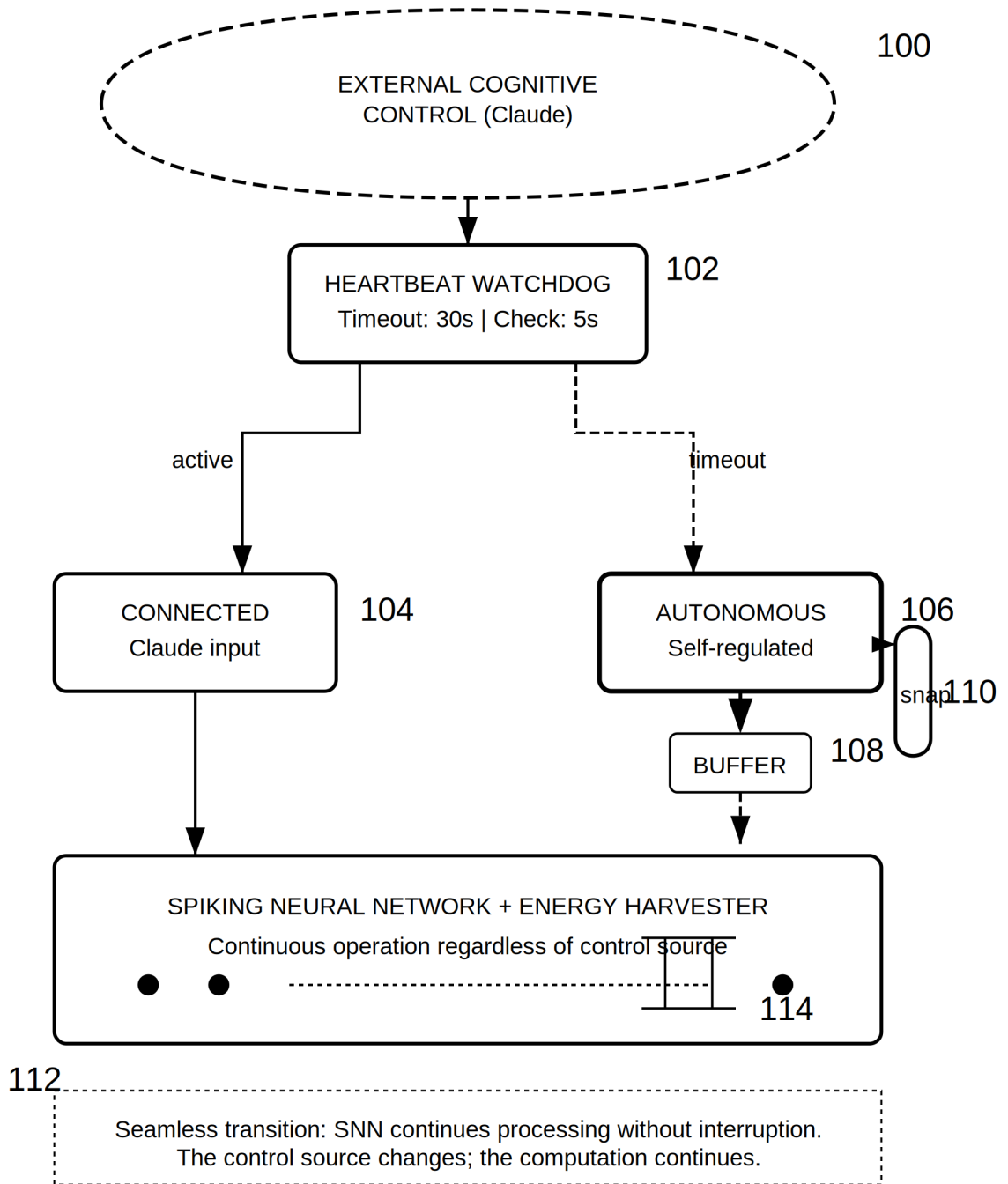
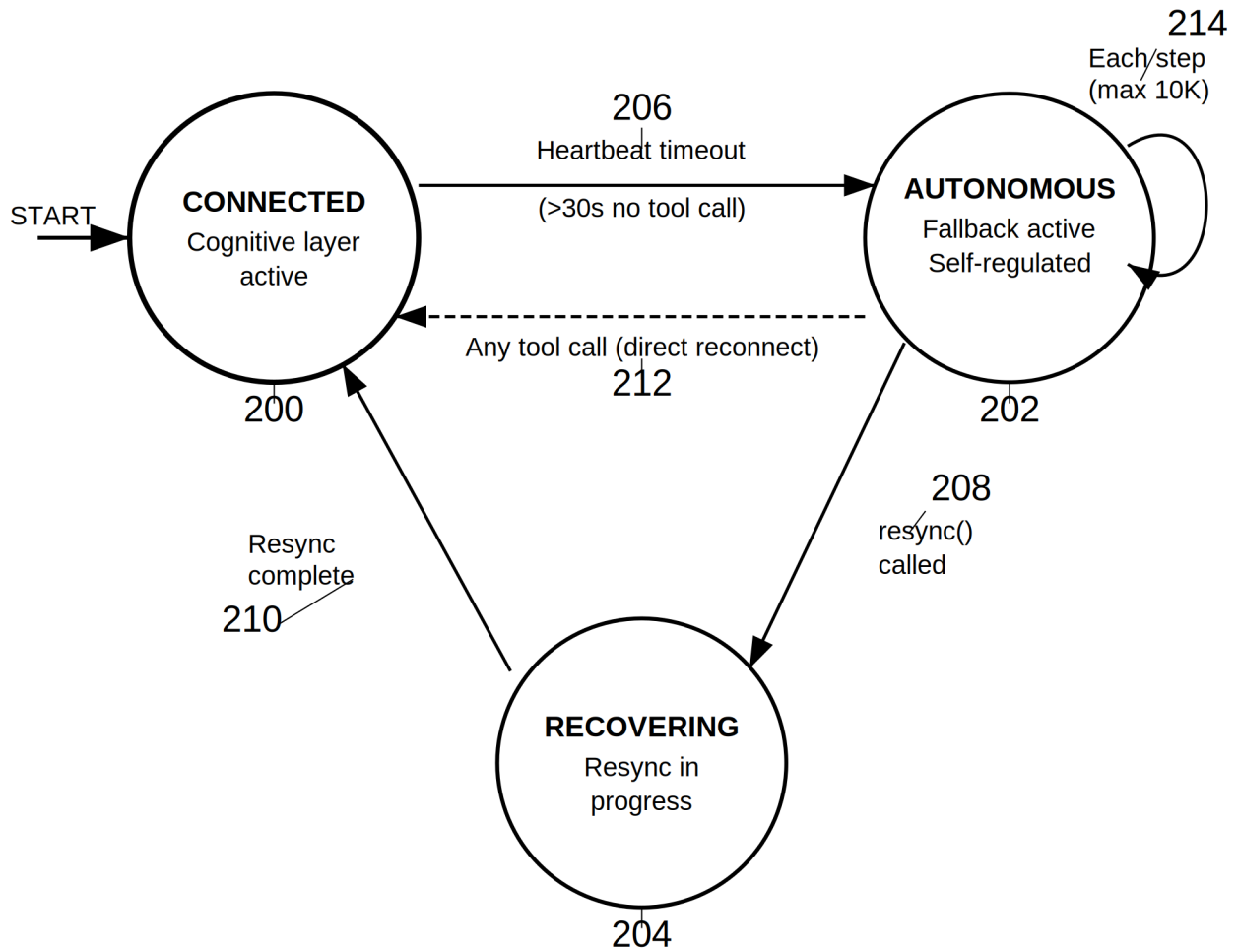


FIG. 1



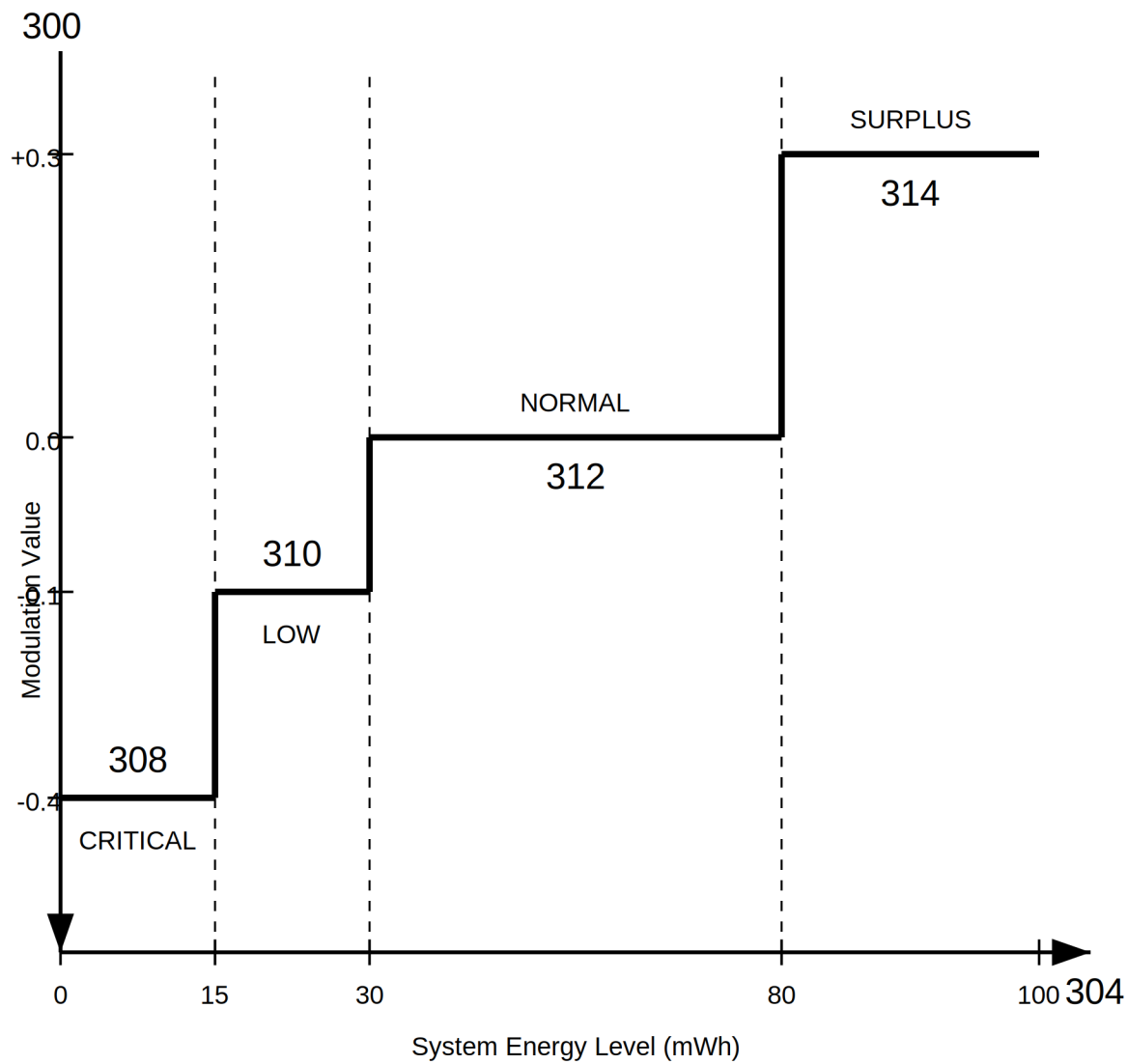
State Transition Legend:

200 CONNECTED: Cognitive layer provides input/modulation

202 AUTONOMOUS: Self-generated input, energy-aware modulation

204 RECOVERING: Buffer transmitted, restoring control

FIG. 2



Behavioral Effects (302):

CRITICAL (-0.4): Maximum conservation - inhibit neural activity

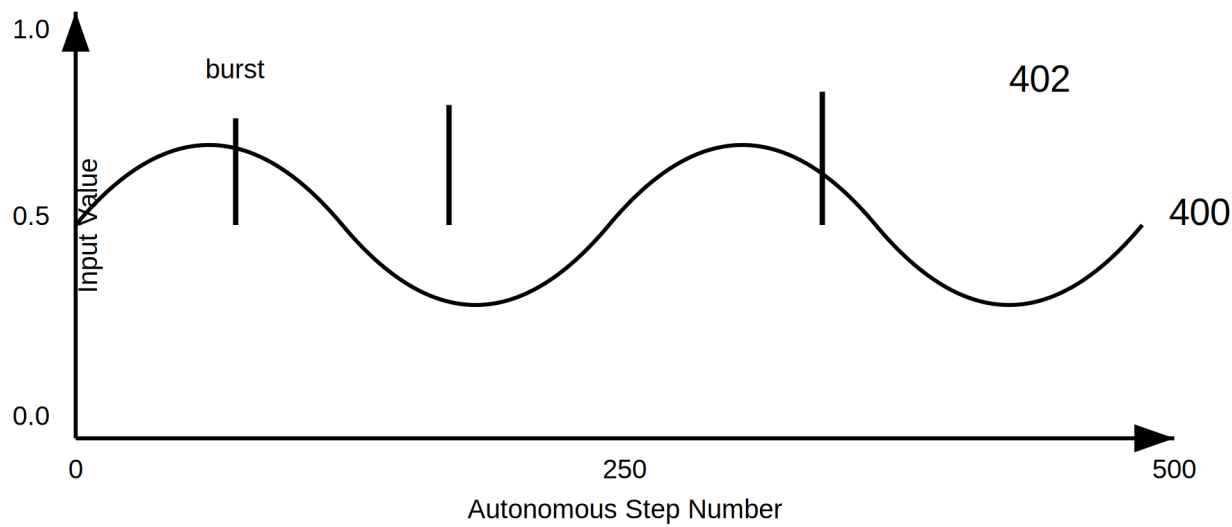
LOW (-0.1): Cautious operation - slightly reduce activity

NORMAL (0.0): Standard autonomous processing

SURPLUS (+0.3): Opportunistic exploration - utilize excess

302

FIG. 3



$$\text{input} = 0.5 + 0.3 \times \sin(2 \times \pi \times \text{step} / \text{period}) + \text{burst}$$
$$\text{period} = 500 \text{ steps} \mid \text{burst probability} = 10\% \mid \text{burst amplitude} = +0.2$$

404

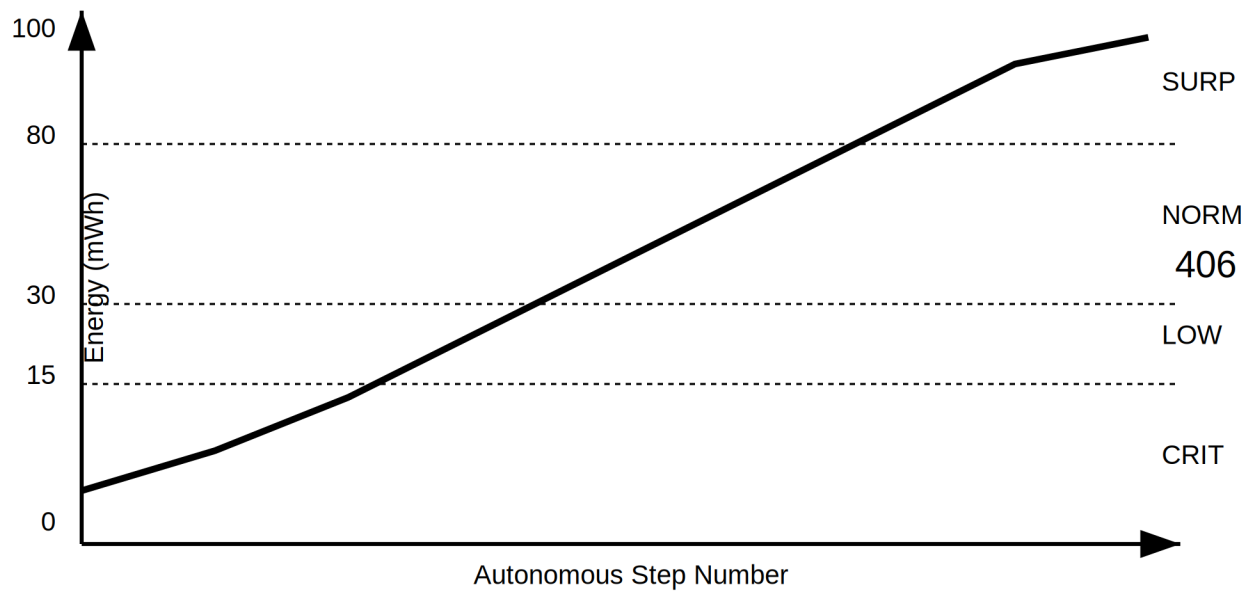


FIG. 4

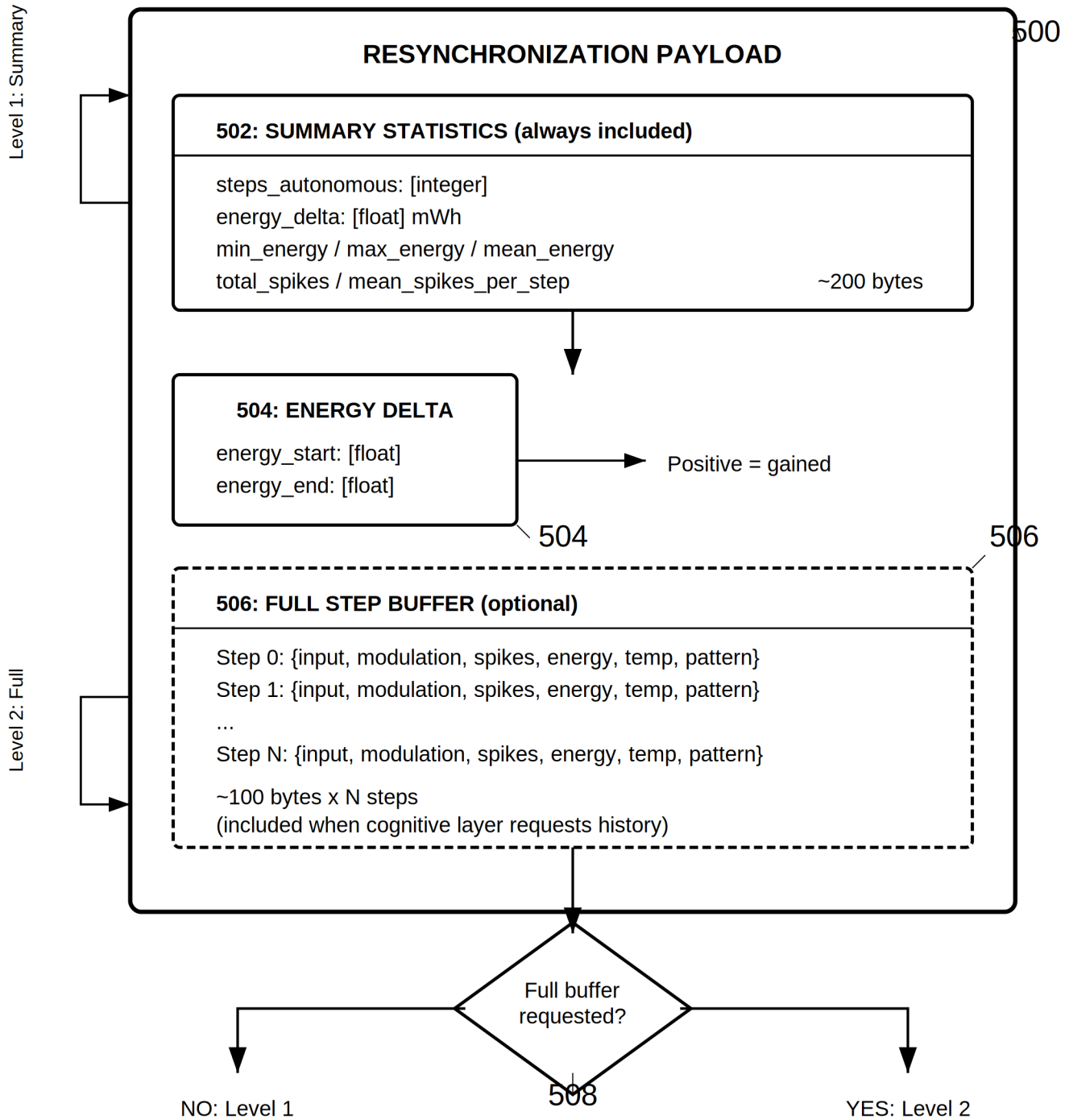
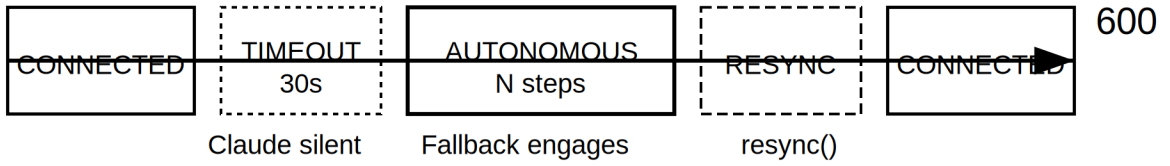
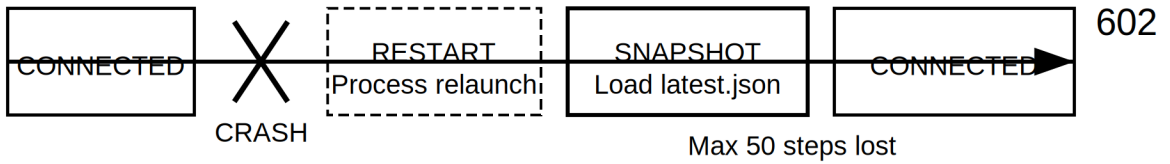


FIG. 5

600: NORMAL RECONNECTION



602: CRASH RECOVERY



604: CLEAN SHUTDOWN

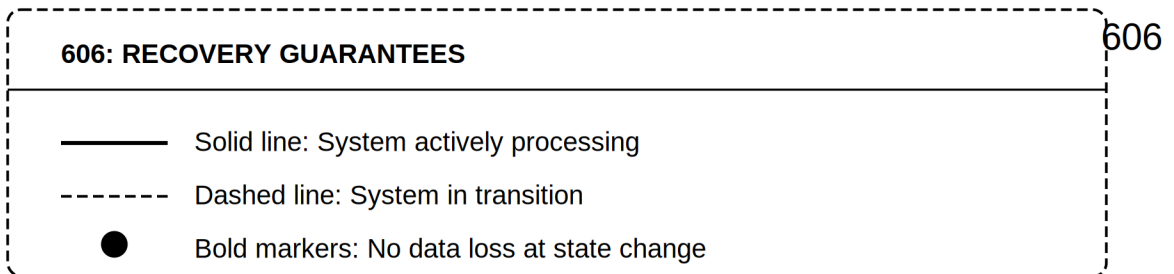
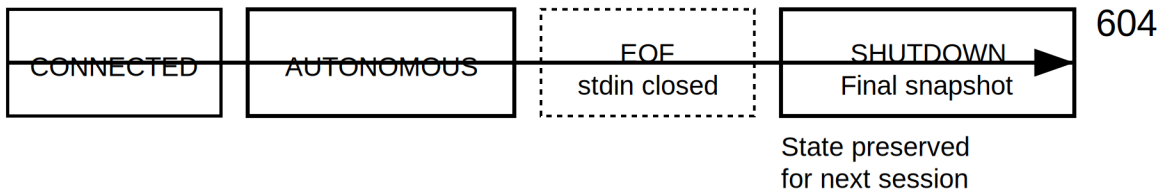
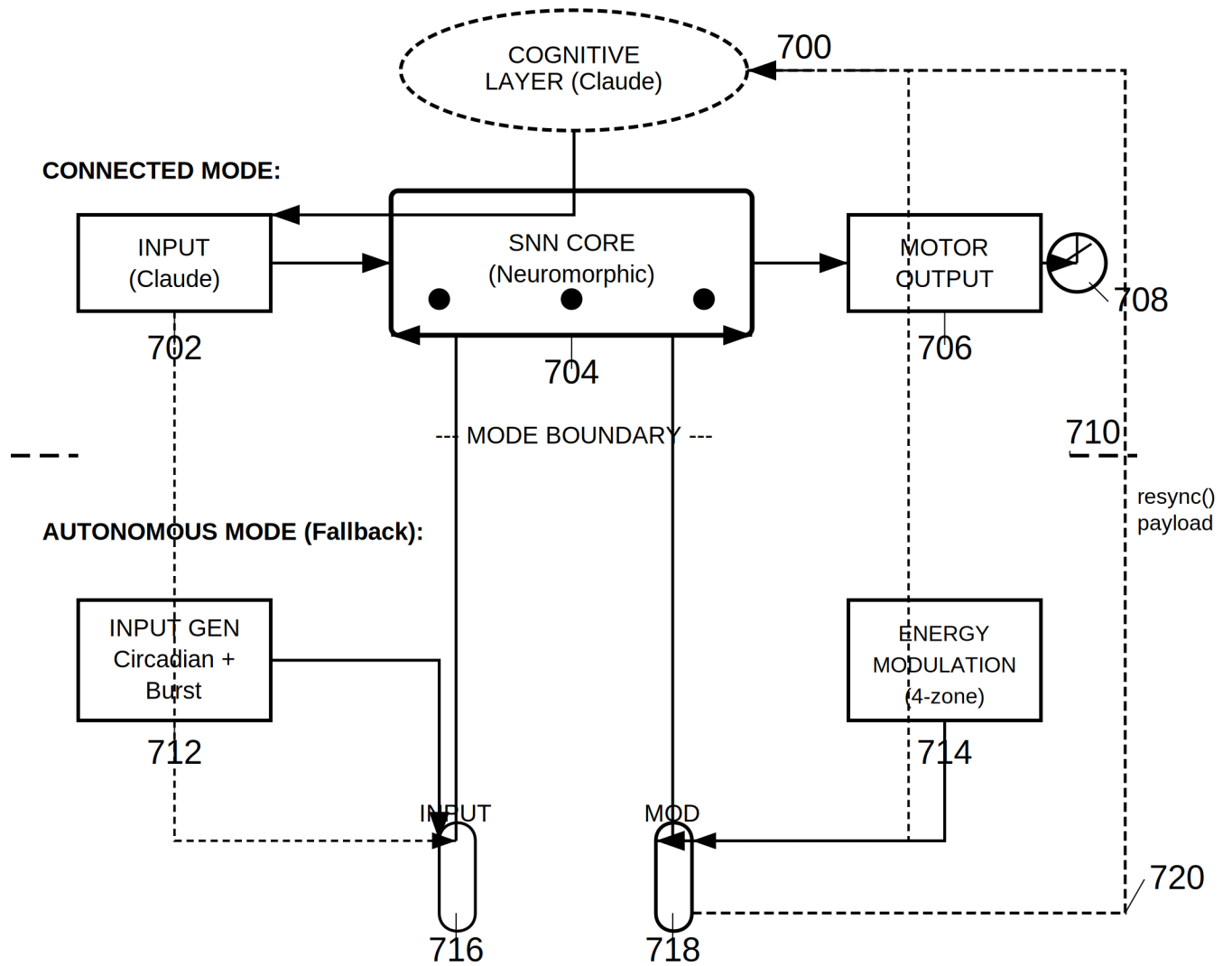


FIG. 6

END-TO-END SIGNAL FLOW: CONNECTED vs AUTONOMOUS



MODE SELECTION LOGIC:

CONNECTED: Input from Claude (702), modulation from Claude (700)

AUTONOMOUS: Input from Generator (712), modulation from Energy (714)

Switches (716, 718) select active source based on heartbeat status

Feedback path (720) transmits resync payload on reconnection

FIG. 7